המרכז
לחינוך סייבר
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# Assembly for Reverse Engineering

## Workspace

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# Objectives

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- Use the install guide to set workspace
- Be familiar with basic assembly file

# Tools

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- Editor: write assembly files
  Notepad++
- Assembler : convert assembly files to machine language
  FASM
- Debugger: examine a program
  IDA
- https://data.cyber.org.il/assembly/32bit/guide.pdf

# Install Guide

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- Use the Install Guide to convert example.asm to example.exe and disassemble it.

C:\Assembly\CE\example.asm - Notepad++

```
1  include 'win32a.inc'
2
3  format PE console
4  entry start
5
6  section '.data' data readable writeable
7      msg1    db  'Welcome to',13,10,0
8      msg2    db  'Cyber Elite assembly class',13,10,0
9
10 section '.text' code readable executable
11 ; ====================================
12
13 start:
14     mov     eax, 0
15     xor     ebx, ebx
16     and     edx, 0
17     mov     ecx, 0x1000
18 again:
19     loop    again
20     ; Show a message to the user:
21     mov     esi, msg1
22     call    print_str
23     ; Show a message to the user:
24     mov     esi, msg2
25     call    print_str
26
27 ; ====================================
28
29     push    0
30     call    [ExitProcess]
31
32 include 'training.inc'
```

CPU - main thread, module example

```
00402000  .  B8 00000000   MOV EAX,0
00402005  .  31DB          XOR EBX,EBX
00402007  .  83E2 00       AND EDX,00000000
0040200A  .  B9 00100000   MOV ECX,1000
0040200F     E2            DB E2
00402010     FE            DB FE
00402011  .  BE 00104000   MOV ESI,OFFSET 00401000        ASCII "Welcome to♪◘"
00402016  .  E8 B9000000   CALL 004020D4                  example.004020D4
0040201B  .  BE 0D104000   MOV ESI,OFFSET 0040100D        ASCII "Cyber Elite assembly c
00402020  .  E8 AF000000   CALL 004020D4                  example.004020D4
00402025  .  6A 00         PUSH 0                         ExitCode = 0
00402027  .  FF15 6830400  CALL DWORD PTR DS:[<&kernel32.ExitProce  KERNEL32.ExitProcess
0040202D  .  55            PUSH EBP
0040202E  .  89E5          MOV EBP,ESP
```

# Basic assembly file

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- While executing, the Operating System reserves memory for the program
- Which parts of a program require memory?
  - Code – the machine language instructions
  - Data – variables (Heap is allocated here)
  - Stack – functions arguments and local variables

# Basic assembly file

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```
include 'win32a.inc'

format PE console
stack 1000h
heap 10000h
entry start

section '.data' data readable writeable
    hi     db  'hi',13,10,0

section '.text' code readable executable
start:

    push    0
    call    [ExitProcess]

include 'training.inc'
```

- The assembly file MUST define text section (code)
- data section – only if there are global variables
- Stack & Heap – OK if not defined, FASM will set defaults

# Basic assembly file

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include 'training.inc'
```

- A variable is defined
- Variable name – “hi”

# Basic assembly file

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```

- Two assembly instructions

# Basic assembly file

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```

- OS should be told how to let the program access memory sections
- What if data memory wouldn't be writable? Will be executable?

# Basic assembly file

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start:

    push    0
    call    [ExitProcess]

include 'training.inc'
```

- Parts covered so far

# Basic assembly file

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    call    [ExitProcess]

include 'training.inc'
```

- PE – Portable Executable. Microsoft format.
- Console – use console (not GUI)

# Basic assembly file

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8  section '.data' data readable writeable
9      hi     db  'hi',13,10,0
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11 section '.text' code readable executable
12 start:
13
14     push    0
15     call    [ExitProcess]
16
17 include 'training.inc'
```

- Entry – where is the 1st instruction?
- Example: You may open a book on page 20
- We can move “start” from line 12 to another line, program execution will start there

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section '.text' code readable executable
start:

    push    0
    call    [ExitProcess]

include 'training.inc'
```

- Include win32a.inc, training.inc– helpful definitions and functions

# Basic assembly file

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    push    0
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```

- Parts covered so far – all ☺